



The NASA Moon Survival Challenge

Overview

The NASA Moon Survival Challenge is a team building activity based on a psychological study carried out by Jay Hall in 1962. The activity has been used in organizations all over the world to build teams and develop team working skills.

However, due to limitations in the instructions that are most readily available, many facilitators miss out on the original purpose of the activity: exploring the impact of training on group effectiveness.

These instructions allow you to run the activity two ways:

1. **As a team building activity:** a variation of the commonly available instructions that use experiential learning review methodology to strengthen learning outcomes
2. **To replicate the original experiment:** this version lets you explore the impact of training on group effectiveness

Time	Level	Participants per team	Materials (per team)	Facilitator materials	Space per team
Briefing: 10m Ind' phase: 5m Team phase: 30m Review: 45 mins Total: 90 mins	Intro	4-6	- List of Items - Paper - Procedural Ground Rules (if running version 2)	PowerPoint Briefing Timer	A small work table

Discussion Topics

- + Fundamentals of effective team working
- + How to make better decisions
- + Exploring the impact of training on group effectiveness (if running version 2)

Preparation

Set out enough Lists of Items on each table for 1 per participant. Set out enough sheets of paper on each table for 1 per participant.



Prepare teams in advance where possible: it's usually a good idea to create teams whose members have a range of interpersonal styles, as this leads typically leads to a richer team dynamic and a more insightful review.

If running version 2, make sure you have enough sets of Procedural Ground Rules for 1 per experimental group (i.e. half of the teams).

Briefing

Use the PowerPoint slides to brief the teams.

Ensure you brief them on the individual stage and the group stage.

- + **Individual:** You must create your own ranking of the items
- + **Group:** You must collectively arrive at consensus for a ranking of the items

If you're running version 2 of the activity, use these briefings for the control and experimental groups:

- + **Control:** Your task, in your discussion groups, is to arrive at a group decision on the NASA task. You must report back to your facilitator when you are satisfied with your answer.
- + **Experimental:** See additional document – [NASA Challenge Experimental Briefing]

Top Tips

- + Just read the brief as is, and don't elaborate.
- + Resist the urge to ask if there are any questions at the end of the briefing, and instead keep the energy high by starting the activity immediately.



During the Task

General

- + Avoid answering questions about the nature of the task: repeating the brief is the best course of action.

During the individual phase:

- + Ensure there is no discussion between participants.
- + Ensure no participants are using their phones.
- + Give a 2 minute warning when the individual phase is nearly finished.

Between phases:

- + If you're running version 2 of the activity:
 - o Split the participants into teams: go for an equal number of control and experimental teams.
 - o Give each experimental team a copy of the Procedural Ground Rules.
- + Ensure each team is on its own table, and has access to 1 List of Items and 1 sheet of paper.

During the team phase:

- + Ensure there is no discussion between teams.
- + Ensure no participants are using their phones.
- + Ensure teams reach consensus before their rankings are collected.
- + Give a 2 minute warning when the team phase is nearly finished.



After the team phase:

- + Go through the correct ranking and get each team to compare the accuracy of their responses to the correct answer (see briefing slides).
 - For this we recommend subtracting the correct ranking for each item from the ranking it was given by a team, for example:
 - If they ranked the tanks of oxygen #3 and the correct answer is #1, their score for this item would be $3-1 = 2$
 - If they ranked the box of matches #1 and the correct answer is #15, their score for this item would be $1-15 = 14$ (**note:** for this method to work, change any negative values to positive)
 - Add up all of a team's item scores
 - The **lowest** total score is closest to the correct answer, and vice versa
- + Get each team to present their results
- + If running version 2 of the activity:
 - Ask participants if they think there's a difference in how experimental and control groups worked
 - Explain the difference between experimental and control
 - See whether control groups or experimental groups performed better

Review phase

- + Give participants review sheets to fill in
- + Discuss thoughts as a group



The NASA Moon Survival Challenge – Experimental Briefing

This is an exercise in group decision making, although there is an individual component to begin.

For the individual component, you must rank the 15 survival items in your perceived order of importance.

Then you will be put into a group, and your group must reach consensus in a shared ranking of items. This means that the ranking for each of the 15 survival items must be agreed upon by each group member before it becomes a part of the group decision.

Consensus is difficult to reach. Therefore, not every ranking will meet with everyone's complete approval. Unanimity, however, is not a goal (although it may be achieved unintentionally) and it is not necessary that every person be as satisfied as he might be, for example, if they had complete control over what the group decides.

What should be stressed is the individual's ability to accept a given ranking on the basis of logic – whatever his level of satisfaction – and their willingness to entertain such a judgment as feasible.

When the point is reached at which a group members feel this way as a minimal criterion you may assume that you have reached a consensus as it is defined here and the judgment may be entered as a group decision.

This means, in effect, that a single person can block the group if he thinks it necessary; at the same time, it is assumed that this option will be employed in the best sense of reciprocity. Here are some guidelines to use in achieving consensus:

1. Avoid arguing for your own rankings. Present your position as lucidly and logically as possible, but consider seriously the reactions of the group in any subsequent presentations of the same point.
2. Avoid 'win-lose' stalemates in the discussion of rankings. Discard the notion that someone must win and someone must lose in the discussion; when impasses occur, look for the next most acceptable alternative for both parties.
3. Avoid changing your mind only in order to avoid conflict and to reach agreement and harmony. Withstand pressures to yield which have no objective or logically sound foundation. Strive for enlightened flexibility; avoid outright capitulation.
4. Avoid conflict-reducing techniques such as the majority vote, averaging, bargaining, coin flipping, and the like. Treat differences of opinion as indicative of an incomplete sharing of



relevant information on someone's part and press for additional sharing, either about task or emotional data, where it seems in order.

5. View differences of opinion as both natural and helpful rather than as a hindrance in decision making. Generally, the more ideas expressed the greater the likelihood of conflict will be; but the richer the array of resources will be as well.
6. View initial agreement as suspect. Explore the reasons underlying apparent agreements; make sure that people have arrived at similar solutions for either the same basic reasons or for complementary reasons before incorporating such solutions in the group decision.



Your involvement in NASA Moon Survival should have stimulated thought about effective teamwork and decision making. To help crystallise your thoughts on the activity please answer the questions below.



My Performance?

	Strongly Disagree	Disagree	Agree	Strongly Agree
I argued for my own rankings				
I took time to understand others' rankings				
I felt I'd 'lost' if my ranking was not accepted				
I wanted to avoid conflict				
I thought differences of opinion were helpful				
I fully understood the rationale behind the team ranking				
I encouraged others to share their ideas				
I helped others understand the rationale of the team rank'				
I spoke freely				
I enjoyed the discussions with my team				
I'm happy with the team ranking				



Team Decision Making

How did the team make decisions during the activity?



What actions did your team members take to gain consensus on the ranking? Be specific

What did you do to help build consensus on the ranking? Be specific

What was the impact of your actions on team performance?



Summary Thoughts

What have you learnt about your colleagues through taking part in this activity? Be specific

What have you learnt about yourself through taking part in this activity?

What can you take from this activity to enhance the way that you and your colleagues work together?